Maximum Regions from Straight Lines: An Incremental Proof and an Audit of Its Recorded Verification

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Abstract

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Provisional synthesis from covered evidence only: For n full straight lines in the Euclidean plane, the supplied deductions establish the maximum region count M(n) = 1 + n(n+1)/2, with regions defined as connected components of the complement. The planned paper explains the argument through two separate tasks: a universal upper bound and an explicit construction attaining it. Convexity of arrangement regions makes the incremental count exact: a nonconcurrent added line with m distinct intersections with earlier lines creates m+1 new regions. Consequently, the kth line adds at most k regions, even when parallelism, concurrency, or repeated lines are allowed. The family L_i: y = ix + i^2 attains every incremental bound because its intersections with L_k have distinct coordinates (-(i+k), -ik). The empty arrangement and the values for n = 0,,4 are treated explicitly, without using examples to infer the theorem. The mathematical deductions are distinguished from their verification provenance: the record accepts the argument but also contains conflicting completion labels, and supplies neither computational evidence nor independent formal verification.

About this paper

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1 Introduction

The problem is an extremal question, not a request to count regions for one favorable drawing: determine the largest number of connected components obtainable after removing n full straight lines from the Euclidean plane, for every nonnegative integer n. A satisfactory answer must both exclude larger counts for arbitrary arrangements and exhibit an arrangement achieving the bound. The empty arrangement is part of the domain, not an exception to be omitted. [C1, C2] The central distinction is between the number of earlier lines and the number of distinct intersection points on a new line. Parallel lines, coincident lines, and several earlier lines meeting the new line at one point can reduce the increment. The upper-bound proof must therefore accommodate degeneracies rather than assume the favorable intersection pattern that the construction will later supply. [C4, C5, C6] The evidence supports a deductive mathematical paper with an explicit provenance discussion, not an experimental report. The substantive argument and the record's completion labels must be evaluated separately: acceptance of the displayed deductions does not resolve the conflicting administrative statuses or constitute independent verification. [C12, C13, C14] *Claims discussed:* C1, C2, C4, C5, C6, C12, C13, C14

2 Evidence and scope

Use the recorded strategy, 'Bound the new regions, then seek equality,' as the organizing method. First identify arrangement regions as convex strict-sign cells; next prove the exact increment for a single insertion; then sum a universal bound. Keep this argument logically separate from the construction so that attainability is not smuggled into the upper bound. [C2, C3, C4, C6, C7] Verify the attaining family symbolically using the intersection calculation recorded in the evidence. For each k, check that L_k meets its k-1 predecessors at distinct points and then apply the already established increment lemma. Treat n=0 and the first few values as base-case and arithmetic checks, not as observations supporting extrapolation. [C8, C9, C10, C11] Synthesize the supplied source-linked index while preserving its evidentiary boundaries. Use the artifact's detailed mathematical extracts as the principal support, the strategy records for the proof architecture, and the session extracts for execution and status information. Report mathematical deductions, reported reviewer assessments, and unresolved record discrepancies as different kinds of claims. No numerical experiment or independent proof check is presented as having been run. [C12, C13, C14] *Claims discussed:* C2, C3, C4, C6, C7, C8, C9, C10, C11, C12, C13, C14

3 1. The Extremal Question and the Scope of the Evidence

The extremal question concerns n full straight lines in the Euclidean plane, where n is a nonnegative integer. Here a line is understood in its entirety, rather than as a segment or ray. A region is a

connected component of the complement of the union of the lines. Thus the objects being counted are the connected portions of the plane remaining after the lines themselves have been removed. The case n = 0 belongs to the same formulation: the union of an empty collection of lines is empty, so its complement is the whole plane and consists of one region. This initial case supplies the starting count for the incremental argument. The objective is to determine the maximum number of regions over all permitted placements of the n lines. Establishing such a maximum requires two logically distinct results. First, a universal upper bound must show that no arrangement produces more than the proposed number of regions. Second, an attaining example must show that the bound can actually be reached for every permitted n. An arrangement producing many regions does not by itself rule out a better arrangement, while an upper bound alone does not establish attainability. The recorded strategy explicitly separates these obligations, treating the construction as the sharpness argument rather than as a substitute for the universal bound. The domain does not require the lines to satisfy an additional distinctness or general-position hypothesis. In particular, the recorded upper-bound argument permits repeated lines as well as parallel lines and concurrent intersections. Its scope is therefore broader than that of a construction selected to attain equality. This distinction matters to the extremal question: favorable intersection patterns may be used to demonstrate attainability, but they cannot be assumed when bounding all arrangements. The empty arrangement must likewise be included explicitly, rather than left implicit in an argument framed only in terms of adding a first line. The proof strategy is incremental. It examines the pieces into which the preceding lines divide an added line and uses that analysis to control the increase in the number of regions. The universal estimate is then paired with a construction whose intersection pattern realizes the required increments. This is a deductive study, not an empirical experiment: the general conclusion rests on the geometric argument and its attaining example. Small cases have a narrower role. They provide checks on the proposed count and its treatment of the initial cases, but they do not establish the result for arbitrary n. The recorded strategy preserves this distinction between checking particular values and proving a universal statement. The mathematical argument must also be distinguished from the history of its recorded assessment. The supplied record contains no computational evidence or external formal verification, and Scientific Python and SageMath were recorded as unavailable. Accordingly, the discussion does not present numerical experiments, software execution, or machine-checked proof as support for the conclusion. This limitation concerns validation provenance; it is not itself a mathematical objection to the deductions. The detailed assessment of the records verification status is reserved for the final substantive section, where reported acceptance of reasoning can be considered separately from claims of independent verification. The evidentiary basis here is a source-linked synthesis of the supplied notes, not an exhaustive review of underlying originals. The evidence package includes summaries, several visibly truncated, and repeated descriptions associated with the same session and artifact. Such repetition cannot be counted as independent corroboration. Moreover, an early extracts statement that artifact contents were absent cannot be generalized to the entire package, because later artifact-linked extracts supply substantial mathematical content. Within these limits, the section establishes the problems domain and the two-part structure of its solution, while keeping the deductive argument distinct from what the available record can substantiate about its verification. *Claims discussed:* C1, C2, C12, C14

4 2. Why Arrangement Regions Are Convex

Let L, , L be the existing full straight lines in the Euclidean plane, where n is a nonnegative integer. A region is a connected component of the complement U = š (L ⋯ L). To identify these components without relying on a drawing, choose an affine equation f(x, y) = ax + by + c for each line L, with (a, b) ≠ (0, 0), so that L is precisely the set where f vanishes. The inequalities f > 0 and f < 0 describe the two open half-planes bounded by L. Every point of U therefore has a well-defined positive or negative sign for each equation. For a sign choice s = (s, , s), with each s equal to +1 or -1, define its strict-sign cell by C = {p š : sf(p) > 0 for every i}. Thus a cell is the intersection of one selected open half-plane for each existing line. Some sign choices may yield an empty intersection; only nonempty cells can represent regions. Each nonempty cell lies in U, because none of its points satisfies any line equation. Conversely, every point of U belongs to exactly one such cell, determined by the signs of its affine equation values. These cells are open and convex. Openness follows because each strict affine inequality defines an open half-plane and there are only finitely many inequalities. For convexity, take p and q in the same cell and any t with 0 ≤ t ≤ 1. Affineness gives sf((1 - t)p + tq) = (1 - t)sf(p) + tsf(q) > 0 for every i. The entire segment joining p to q therefore remains in the cell. In particular, a nonempty cell is connected: any two of its points are joined within it by that segment. It remains to establish that these cells are whole regions, rather than merely connected subsets of regions. Let R be a connected component of U. For each i, the function f is nonzero throughout R and cannot take both positive and negative values there. Indeed, if it did, the sets R {f > 0} and R {f < 0} would be disjoint, nonempty, relatively open subsets whose union is R, contradicting connectedness. Consequently, every equation has a fixed sign on R, and R is contained in the cell specified by those signs. This is the precise connectedness argument behind the assertion that a region cannot cross a removed line. Conversely, let C be a nonempty cell and choose a point p in it. Since C is connected and contained in U, it lies in the connected component R containing p. The preceding argument shows that all points of R have the same signs as p, so R is contained in C as well. Hence R = C. The regions of the arrangement are therefore exactly the nonempty strict-sign cells, and every region is open and convex. No assumption of distinctness or general position is needed for this identification. Repeated line equations, parallel lines, or several lines through one point may affect which sign choices are nonempty, but not the argument identifying the nonempty cells with the components. The empty arrangement fits the same description. When n = 0, there are no inequalities to impose, so the single empty sign choice specifies all of š. With no lines removed, the plane is one connected region. This supplies the initial case while preserving the same interpretation of regions as nonempty intersections of chosen half-planes. Openness and convexity provide the structural facts needed when another line is inserted. Convexity ensures that the intersection of an old region with the new line is connected whenever it is nonempty: the segment between any two intersection points stays in the region. Thus separated portions of that line cannot leave an old region and later return to it across an excluded point. Openness supplies the complementary local fact. If the new line passes through a point of an old region, a sufficiently small open disk around that point remains in the region and contains points on both sides of the line. Intersecting the region with either of the new lines open half-planes then gives a nonempty convex set. These properties, rather than the appearance of a particular drawing, justify the subsequent analysis of how inserting a line divides the existing regions. *Claims discussed:* C1, C3

5 3. The Exact Increment Produced by One New Line

Let a finite collection of existing full straight lines be given, and let its regions be the connected components of the plane after those lines are removed. The exact increment statement is as follows: if a new line L is not coincident with any existing line and meets their union at m distinct points, then inserting L increases the number of regions by exactly m + 1. Establishing this statement requires more than counting pieces of L. One must show that those pieces lie in different old regions and that each such region is divided into exactly two new regions. The openness and convexity of the old regions provide these two bridges. To make those properties explicit, write the existing lines as the zero sets of affine equations f, , f. At every point in their complement, each f has either positive or negative sign. On a connected component of the complement, that sign cannot change: the positive and negative subsets would otherwise separate the component into two nonempty relatively open sets. Thus every old region lies in a strict-sign cell, an intersection obtained by choosing either $f > 0$ or $f < 0$ for each existing line. Conversely, every nonempty such cell is an intersection of open half-planes and is therefore open and convex. In particular, it is connected and lies entirely in the complement. It must consequently be contained in a single region. These two inclusions identify the regions precisely with the nonempty strict-sign cells, proving the openness and convexity needed below. Because L is noncoincident with every existing line, its intersection with each predecessor is either empty or a single point. The union of these intersections therefore consists of finitely many distinct points. When $m > 0$, order those points along L. Removing them leaves two unbounded end pieces and the m - 1 open segments between successive points, for a total of m + 1 connected pieces. When $m = 0$, nothing is removed from L, and the whole line is one connected piece, again giving m + 1. Every piece avoids all existing lines and is connected, so it lies within one old region. No two different pieces can lie in the same old region. Suppose, to the contrary, that points p and q from different pieces belonged to one old region R. Since R is convex, the entire segment joining p to q would belong to R. But p and q lie on L in different components of the line with its old intersection points removed, so the segment between them contains at least one of those intersection points. That point lies on an existing line and hence cannot belong to R. The contradiction proves that the m + 1 pieces encounter m + 1 distinct old regions. Moreover, every old region met by L contains a point of one of these pieces, since an old region contains no point on an existing line. Thus these are exactly the regions affected by the insertion. Consider one encountered old region R, and choose a point p in R L. Let g be an affine equation for L, so that its two strict half-planes are $g > 0$ and $g < 0$. The openness of R implies that a sufficiently small open disk centered at p lies inside R. Because p lies on L, that disk contains points on both sides of L. Consequently, both R $\{g > 0\}$ and R $\{g < 0\}$ are nonempty. This is the step that ensures an encountered region is genuinely split: merely knowing that L meets R would not, without the openness argument, establish the presence of region points on both sides. Each of these two intersections is convex, being the intersection of the convex set R with a half-plane, and is therefore connected. Together they exhaust R after L is removed. They cannot belong to one connected component of that remainder, because the positive and negative signs of g separate it into two disjoint relatively open subsets. Hence R is replaced by exactly two connected components, not merely by at least two. These are also components of the complement of the full enlarged arrangement: a connected subset of that complement is contained in the old complement and therefore cannot join points from different old regions. Each encountered old region thus contributes precisely one net additional region. An old region not met by L is unchanged by the insertion. Combining this observation with the preceding count gives the exact increment: there are m + 1 encountered old regions, each is replaced by two regions, and every other old region remains intact. If the old and new region counts are denoted by N_old and

N_new, respectively, then N_new - N_old = m + 1. The argument also covers m = 0 directly: the entire new line lies in one old region, and that region is divided into exactly two. Coincidence must be treated separately from this lemma. If L coincides with an existing line, adding it does not change the union removed from the plane and therefore creates zero new regions. Parallelism and concurrency, by contrast, require no modification of the noncoincident proof. An existing line parallel to L contributes no intersection point, while several existing lines meeting L at the same point contribute only one distinct point to m. The count is therefore a count of distinct locations on the new line, not a count of pairwise incidences. No general-position assumption is needed: for every noncoincident insertion the exact increment remains m + 1, while a coincident insertion has increment zero. *Claims discussed:* C3, C4, C5

6 4. A Universal Upper Bound Without General Position

Fix a nonnegative integer n and an arbitrary ordered list of n full straight lines in the Euclidean plane. No restriction is placed on parallelism, concurrency, or repetition. For each k with $0 \leq k \leq$ n, let A_k be the number of regions after the first k lines have been inserted, where a region is a connected component of the complement of their union. Before any insertion, the complement is the entire plane, so A_0 = 1. The upper-bound argument concerns the increments A_k - A_{k-1} for this particular arrangement; it does not assume that the arrangement maximizes the region count. First consider an insertion in which the new line is not coincident with any existing line. If it meets the existing union at m distinct points, those points divide it into exactly m+1 connected pieces. Each piece avoids the existing lines and therefore lies within an old region. The exact increment lemma gives an increase of m+1 regions, rather than merely an upper estimate. Its geometric basis is the convexity of the old regions: each is a nonempty intersection of open half-planes determined by the existing lines. Two different pieces of the new line cannot lie in the same old region, because the segment joining points from those pieces would contain an intersection with an existing line, contrary to convexity of that region. Each of the m+1 pieces also produces an actual split of its old region. At a point of the piece, openness of the region supplies nearby points on both sides of the new line. Intersecting that region with the two strict half-planes bounded by the new line therefore gives two nonempty convex sets, which are precisely its two replacement regions. Thus every encountered old region contributes one net additional region. Regions not encountered by the new line remain unchanged, establishing the increment m+1. Apply this count to the kth insertion, for $1 \leq k \leq$ n. If the kth line is noncoincident with its predecessors, each of the k-1 preceding lines contributes at most one intersection point on it. Consequently, the number m of distinct intersection points satisfies $m \leq$ k-1, and hence A_k - A_{k-1} = m+1 $\leq$ k. Parallelism and concurrency do not create exceptions to this estimate. A parallel predecessor contributes no intersection point, while several predecessors meeting the new line at a common point contribute only one distinct point there. These possibilities can reduce the distinct-intersection count, but cannot increase it beyond k-1. If instead the kth line coincides with a predecessor, its insertion leaves the union of the lines unchanged and adds zero regions. This case also satisfies A_k - A_{k-1} $\leq$ k, although it must be treated separately from the noncoincident increment lemma. The inequality therefore holds at every insertion in every ordering of the arrangement. Summing it telescopically yields A_n - A_0 = _{k=1}^n (A_k - A_{k-1}) $\leq$ _{k=1}^n k. Using A_0 = 1 and evaluating the triangular sum gives A_n $\leq$ 1 + n(n+1)/2. Since the original arrangement and its ordering were arbitrary, this is a universal upper bound for the number of regions formed by n full straight lines, without any general-position hypothesis. The same expression includes n = 0: the sum then has no terms and

equals zero, leaving the single region of the empty arrangement. The conclusion established here is an upper bound, not an equality recurrence for arbitrary arrangements. Repeated lines have zero increment, and parallelism or concurrency can prevent a noncoincident insertion from reaching its upper bound. Showing that 1 + n(n+1)/2 is attained requires a separate argument that realizes all the necessary increments; it does not follow from summing the inequalities alone. *Claims discussed:* C1, C4, C5, C6, C7

7 5. A Concrete Family Attaining Every Increment

To establish attainability for every n, consider the single infinite family of lines L_i: y = ix + i^2, indexed by the positive integers i. For a given nonnegative integer n, take its first n members; when n = 0, take the empty arrangement. Thus the construction does not require a separate choice of lines for each value of n. Each L_i has slope i, so distinct members have distinct slopes and are therefore neither parallel nor coincident. It remains to show that, when these lines are inserted in index order, each new line meets all its predecessors at distinct points. This supplies the intersection count required by the exact increment lemma. Fix indices i < k. At the intersection of L_i and L_k, their defining equations give ix + i^2 = kx + k^2. Rearranging yields (i - k)x = k^2 - i^2 = (k - i)(k + i). Since i ≠ k, division by i - k gives x = -(i + k). Substituting this value into the equation for L_i gives y = i[-(i + k)] + i^2 = -ik. Consequently, the unique intersection of these two lines is L_i L_k = (-(i + k), -ik). The calculation explicitly identifies where each predecessor meets the newly inserted line, rather than relying only on the fact that their slopes differ. Now hold k fixed and let i range over 1, , k - 1. The x-coordinates -(i + k) are distinct: equality of the coordinates for two indices i and j would imply -(i + k) = -(j + k), and hence i = j. The corresponding intersection points on L_k are therefore all distinct. Since each of the k - 1 preceding lines meets L_k once, these are exactly the k - 1 distinct points at which L_k meets the existing arrangement. Distinct slopes ensure that no predecessor is missed, while the coordinate calculation ensures that different predecessors do not meet L_k at the same point. The exact increment lemma now applies with m = k - 1. Recall that this lemma concerns a noncoincident added line with m distinct intersection points with the existing lines: those points divide it into m + 1 connected pieces, each lying in a different old region. The distinction between pieces and regions is justified by convexity. If two pieces lay in the same old region, the segment joining points on them would have to remain in that region, although it passes through an intervening point on an old line. Moreover, each encountered region is open and convex, so the added line separates it into exactly two nonempty convex parts. Each encountered region therefore contributes one net new region. For L_k, the k - 1 intersection points consequently give exactly k new regions, not merely an upper estimate. The initial insertion is included in the same argument. When k = 1, there are no preceding lines, so m = 0 and L_1 creates exactly one new region. If r_k denotes the number of regions after inserting L_1, , L_k, the empty arrangement gives r_0 = 1, and the construction satisfies r_k - r_{k-1} = k for every k ≥ 1. Thus every insertion has the exact increment needed for attainment; the conclusion is not based on extrapolating a pattern from small arrangements. Summing these exact increments gives r_n = 1 + $\sum$_{k=1}^{n} k = 1 + n(n + 1)/2. For n = 0, the sum is empty and the same expression gives the single region of the plane with no lines removed. Hence every finite prefix of the family has the stated region count. The construction establishes attainability for all nonnegative integers n by an explicit intersection calculation and the increment lemma, without requiring computational checks or a different arrangement for each n. *Claims discussed:* C4, C8, C9

8 6. The Maximum Formula and Its Boundary Checks

Let M(n) denote the maximum number of connected regions in the complement of n full straight lines in the Euclidean plane, where n is a nonnegative integer. The universal upper bound and the explicit attaining construction establish complementary parts of this extremal problem. The upper bound limits every allowable arrangement, including arrangements with parallel, concurrent, or repeated lines; the construction shows that this limit can actually be reached. Combining the two gives M(n) = 1 + n(n+1)/2 for every n ≥ 0. This equality concerns the maximum over arrangements, not the region count of every arrangement. To make the upper-bound implication explicit, let A_k be the number of regions after the first k lines of an arbitrary arrangement have been included. The empty arrangement has A_0 = 1. The incremental argument gives A_k - A_{k-1} ≤ k: a noncoincident kth line has at most k-1 distinct intersections with preceding lines and therefore creates at most k new regions, while a coincident line creates none. Summing these inequalities yields A_n ≤ 1 + ∑_{k=1}^n k = 1 + n(n+1)/2. Because the arrangement was arbitrary, this proves M(n) ≤ 1 + n(n+1)/2. Nothing in this step requires all the incremental bounds to be attained simultaneously. The family L_i: y = ix + i^2 supplies that simultaneous attainment. For each k ≥ 1 and each predecessor i < k, the intersection of L_i with L_k is (-(i+k), -ik). With k fixed, different values of i give different x-coordinates, so the kth line meets its k-1 predecessors at exactly k-1 distinct points. The exact increment result therefore gives k new regions at step k. This also covers k = 1, when there are no preceding intersections and the first line creates one additional region. Consequently, the first n lines of this family produce exactly 1 + ∑_{k=1}^n k = 1 + n(n+1)/2 regions. They provide the lower bound M(n) ≥ 1 + n(n+1)/2, matching the universal upper bound and completing the extremal conclusion. The boundary case n = 0 is included directly rather than supplied by extrapolation. With no lines removed, the plane is one region, so M(0) = 1. The sum of increments is then empty, and the formula likewise gives 1 + 0(0+1)/2 = 1. The attaining family at n = 0 is simply its empty prefix. Thus both the counting argument and the construction have the same initial condition, and neither requires an exception to the stated domain n ≥ 0. The recorded small-value checks are M(0), M(1), M(2), M(3), M(4) = 1, 2, 4, 7, 11. Starting from one region, these values correspond to successive additions of 1, 2, 3, and 4 regions in the attaining construction. Substitution into the formula gives the same sequence. Their role is arithmetic consistency: they check the initial count and the first few increments against the general expression. They are not used to infer the formula from a numerical pattern. The conclusion for every nonnegative integer n rests instead on the universal inequality and the construction attaining equality for each finite prefix. These checks must also be distinguished from executed experiments. The supplied record contains no computational evidence or external formal verification; Scientific Python and SageMath were recorded as unavailable, and no persisted executions are supplied. Although the displayed deductive steps were accepted and the review described the argument as convincing, that assessment was explicitly unverified. The record also contains unexplained differences between acceptance of the argument and incomplete, partial, or unresolved status labels. Accordingly, the maximum formula is supported here by the stated deductions, not by computational confirmation or an independently verified execution record. Those provenance limitations qualify claims about verification without supplying a mathematical counterexample to the matching-bound argument. *Claims discussed:* C7, C9, C10, C11, C12

9 7. What the Record Establishesand What Its Status Labels Do Not

The supplied record requires a distinction between the mathematical argument, the reviewers reported assessments of that argument, and the metadata describing completion and execution. These layers do not yield a single consistent account of verification. The session and its linked cycle-1 artifact supply a deductive upper-bound argument and an attaining construction, and their review fields report acceptance of all six reasoning steps. Other fields nevertheless describe the work as partial, unresolved, or incomplete. The appropriate conclusion is therefore neither that favorable review labels constitute independent verification nor that unfavorable completion labels identify a mathematical defect. The conflict is a limitation of the recorded assessment whose reconciliation remains unresolved (C13). The positive assessments are substantive rather than merely generic expressions of approval. The supplied extracts report acceptance of the argument identifying arrangement regions with nonempty strict-sign cells, establishing their convexity, and determining the exact increment produced by an added line. They also report acceptance of the universal upper bound, the construction attaining it, and the treatment of the empty arrangement and small cases. In the recorded argument, a noncoincident line meeting the preceding lines at m distinct points adds exactly m+1 regions, whereas a coincident line adds none. The kth line consequently adds at most k regions. The construction y = ix + i^2 supplies the matching increments through its distinct intersections with preceding lines. These are the mathematical contents to which the favorable assessments refer; acceptance is reported for all six structured reasoning steps, not merely for the final formula. The accompanying closure language is similarly explicit. The investigator describes the argument as complete and self-contained and reports no open obligations. The skeptic calls it convincing, records no material objection, and supplies an empty objections list. Further review fields describe the objective as achieved, the claim as unconditional, and the strategy as a solution, with no remaining proof obligations or unresolved objections. The artifact also records acceptance of global closure on the stated basis that the objective was achieved and no obligations or objections remained. These statements establish what the review process reported, but their evidentiary force must remain distinct from independent verification. The same material retains the designation convincing_unverified and does not claim an external formal check (C12C13). Against these favorable assessments, the supplied record contains incompatible overall status descriptions. Session extracts characterize the outcome as mixed or partial progress and the verification as incomplete; another caveat says that the saved argument remains incomplete and that no solution is claimed. The artifact likewise combines accepted reasoning steps and accepted closure with an objective-review status of partial. A review response reports no material objection while recording a stop reason of unresolved. At the execution level, the session is marked ended or Completed, but its stopping reason is Stop no progress and its last decision is stop_no_progress. The supplied fields do not explain how these descriptions are to be reconciled. Selecting either the favorable closure statements or the unfavorable status labels as the definitive global assessment would therefore go beyond the record (C13). The obligation accounting exposes the discrepancy in a more specific form. One session summary counts five proof obligations as open, although its two displayed obligation entries describe region identification and convexity, and the exact increment for an added line, as established. The other three entries are not shown in that summary. Artifact-linked extracts, by contrast, report all five obligations as established and identify their subjects as region identification and convexity, exact increments, the universal upper bound, an attaining construction, and the empty-arrangement and small-value checks. Those artifact descriptions can be reported, but they should not be substituted for the missing text of the three undisplayed session entries. The available material supports a discrepancy between counts and reported obligation states, not an explanation of how that dis-

crepancy arose or a reconstruction of the absent entries (C13). The execution evidence imposes a separate limitation. Scientific Python and SageMath were both recorded as unavailable. The extracts report null Python and Sage code or computational feedback, empty computational citations, no persisted executions, and recorded execution counts of zero. Listed tool configurations therefore do not establish that a computation took place. The supplied record contains neither computational evidence for the result nor external formal verification of the proof. In particular, the reported values 1, 2, 4, 7, and 11 for the empty arrangement and the next four cases are presented as deductive checks, not as outputs from an executed program or as evidence from which the general theorem was inferred (C12). This absence of execution should not be converted into either an additional endorsement or a mathematical objection. The stated method is deductive, with the universal upper bound established separately from the construction attaining it, and the record describes computation as neither needed nor available. Unavailable tools do not independently confirm any step of that reasoning, but neither do their unavailability or the absence of execution records contradict the deductions supplied. Likewise, the reviewers agreement is evidence of a reported assessment rather than an independent computational or formal validation. The narrower conclusion supported by the verification record is that the argument was presented and favorably reviewed within the recorded process, while independent verification was not supplied (C12C13). The provenance of the supplied material further limits the scope of this audit. The evidence index identifies its extracts as untrusted summaries of saved originals rather than original verification, and several entries are visibly truncated. Repeated descriptions of the same session and artifact do not constitute independent replications or corroboration by multiple sources. There is also a local difference in coverage: an early session extract says that the logged artifacts contents are not supplied, whereas later artifact-linked extracts provide substantial mathematical and review content. The earlier absence should not be generalized to the entire package, but the later material does not justify a claim that the underlying originals have been exhaustively reviewed. The source-linked synthesis rests on the supplied session and artifact extracts, with those coverage limitations retained (C14). Taken together, the session record, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa, and its linked artifact, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, support a qualified account of the recorded verification. They contain a deductive argument, acceptance of six reasoning steps, and reports of no remaining objections or obligations. They also contain incompatible completion labels and obligation counts, without an explanation resolving them. No computational execution or external formal verification supplies an additional basis for adjudication. The mathematical reasoning can therefore be discussed on its stated deductive grounds, while the provenance and status of its recorded review remain explicitly limited and internally inconsistent (C12C14). *Claims discussed:* C12, C13, C14

10 8. Conclusions and Remaining Validation Work

The deductively supported conclusion is that the maximum number of regions formed by n full straight lines in the Euclidean plane is M(n) = 1 + n(n+1)/2 for every nonnegative integer n, where regions are the connected components of the complement of the lines. The conclusion rests on two distinct parts: a bound applying to every allowable arrangement and a construction attaining that bound for every n. Keeping these parts separate is essential to the recorded argument. An arrangement with many regions would not alone establish maximality, while an upper bound would not alone establish attainability. The supplied deductions provide both, including the empty arrangement, for which M(0) = 1 (C2, C10). The central methodological step is the exact local

increment. In the recorded proof, the regions of an existing arrangement are nonempty intersections of open half-planes and are therefore open and convex. A noncoincident added line meeting the old lines at m distinct points is divided into m+1 pieces. These pieces lie in different old regions, and each encountered region is split into exactly two, so the increase is exactly m+1. A coincident line instead adds no regions. For the kth line, there can be at most k-1 distinct intersection points with its predecessors, giving an increase of at most k. Summing these bounds from the initial single region yields 1 + n(n+1)/2. Parallelism, concurrency, and repeated lines do not require exclusion from this universal argument: they cannot increase the relevant distinct-intersection count (C2, C10). Equality is established separately by the family L: y = ix + i^2, for i = 1, , n. For i < k, the intersection of L and L is (-(i+k), -ik). For fixed k, these points are distinct as i varies, so the kth line meets its k-1 predecessors at k-1 distinct points and creates exactly k new regions. Thus every incremental upper bound is attained by the same family, and its first n lines attain the claimed total for every positive n. Together with the empty case, this matches the universal bound throughout the stated domain. The reusable structure of the proof is consequently to establish an exact local change, bound that change universally, and then verify equality through an explicit construction, rather than to assume an extremal configuration when proving the bound (C2, C10). The recorded values M(0), , M(4) = 1, 2, 4, 7, 11 are consistent with the initial region count and successive increments 1, 2, 3, and 4. Their role is nevertheless limited to checking the argument in small cases. They are not the basis for extrapolating the quadratic formula, and the supplied record does not present them as a computational experiment. Generating further small examples would not replace either the universal upper-bound argument or the attaining construction. The general conclusion depends on those deductions, not on the length of an observed sequence (C2, C11). The mathematical conclusion must also be distinguished from the provenance of its recorded validation. The supplied extracts report acceptance of all six reasoning steps and describe the argument as convincing, but they do not supply independent proof verification or external formal verification. Scientific Python and SageMath were recorded as unavailable; code and computational feedback were null, computational citations were empty, and no persisted executions were reported. These limitations prevent a claim that computation or machine checking corroborated the result. They do not, by themselves, identify a defect in the deductive chain. On the supplied mathematical account, no specific unresolved mathematical step is identified, although the record does not establish independent verification of that account (C10, C12, C13). Administrative closure is less clear than the displayed mathematical reasoning. Some fields describe the objective as achieved, accept the reasoning steps, and report no remaining proof obligations or unresolved objections. Other fields classify the outcome as partial, mixed, unresolved, or incomplete, and the coordinator decision is recorded as stop_no_progress. The obligation accounting is likewise inconsistent: a session summary counts five open obligations, while displayed obligation entries describe their subjects as established and artifact extracts report all five established. The supplied evidence does not explain these discrepancies. It would therefore be unwarranted either to select the most favorable completion label as definitive or to interpret an unexplained incomplete label as a specific mathematical counterargument (C13). The scope of the evidence review imposes a further qualification. The available material consists of supplied summaries and extracts, several visibly truncated, rather than an exhaustive examination of the underlying originals. Repeated descriptions of the same session and artifact provide source-linked accounts, not independent replications. An early extract's statement that artifact contents were absent must also be read locally, since later artifact-linked extracts contain substantial mathematical detail. The appropriate conclusion is thus a synthesis of the supplied deductive evidence, with its provenance limits preserved, rather than a claim that every original record has been inspected or that multiple independent sources have verified the theorem (C14). The remaining validation work should therefore be directed toward two separate tasks:

reconciling the completion and obligation fields against the underlying records, and obtaining an independent assessment of the mathematical chain. The first would clarify administrative status; the second would address the explicitly unverified provenance of the proof. Neither task is equivalent to producing more small cases, and neither should be represented as already completed. The final position is correspondingly precise: the supplied deductions support the exact maximum M(n) = 1 + n(n+1)/2 and identify no specific outstanding mathematical gap, while the record's administrative reconciliation and independent verification remain unsettled (C10C14). *Claims discussed:* C2, C10, C11, C12, C13, C14

11 Verification status and limitations

- IMPORTANT EVIDENCE COVERAGE LIMITATION: 1068 saved research field fragment(s) were omitted during indexing and were not reviewed for this paper. The synthesis below is provisional and reflects covered evidence only; no conclusion is drawn from omitted material or its absence. Consult the complete frozen evidence snapshot and omission ledger before relying on these findings.
- The evidence is a bounded, source-linked extraction. Several entries are truncated, and repeated summaries of one session or artifact do not provide independent corroboration. [C14]
- The argument is supported as a deduction in the supplied record, but no external formal or independent proof verification is supplied. Reviewer agreement is not a substitute for such verification. [C10, C12]
- Scientific Python and SageMath were unavailable, and no computational evidence is reported. The small-value table must not be described as the result of executed code. [C11, C12]
- Completion and proof-obligation labels conflict across the session and artifact extracts. The plan cannot substantiate an unqualified claim that the recorded workflow closed successfully. [C13]
- The theorem concerns full straight lines in the Euclidean plane. The supplied argument does not establish analogous formulas for segments, curves, bounded domains, or higher-dimensional arrangements. [C1, C14]

12 Open obligations

- Reconcile the achieved/no-remaining-obligations assessments with the partial, incomplete, unresolved, and stop_no_progress fields. This is a record-consistency obligation, not an identified counterexample to the theorem. [C13]
- Recover the complete obligation register, including the three entries not displayed in the session summary, and determine why five obligations are counted open while other extracts describe all five as established. Do not infer the missing session entries solely from artifact summaries. [C13, C14]

- If independent verification is required, obtain and review the underlying originals or commission an explicit independent proof check. Neither activity is documented as completed in the supplied evidence. [C12, C14]
- No specific unresolved mathematical lemma is identified by the supplied detailed deductions. Preserve that narrower observation without treating it as a resolution of the conflicting global completion statuses. [C3, C4, C7, C9, C13]

13 Frozen obligation reconciliation

Each item is compared with the complete frozen evidence snapshot. An item not listed by the author is not presumed resolved. Source references identify the saved record and field.

Status and source	Obligation
Not listed by author; not presumed resolved; session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:proof_obligations, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f:proof_obligations	Convexity and identification of arrangement regions: established by reasoning step 1 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.
Not listed by author; not presumed resolved; session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:proof_obligations, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f:proof_obligations	Exact increment m+1 for a noncoincident line and zero for a coincident line: established by reasoning step 2 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.
Not listed by author; not presumed resolved; session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:proof_obligations, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f:proof_obligations	Universal upper bound for arbitrary arrangements: established by reasoning step 3 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.

Not listed by author; not presumed resolved; session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:proof_obligations, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f:proof_obligations	Existence of an attaining arrangement for every positive integer n: established by the intersection calculation and increment argument in reasoning steps 45 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.
Not listed by author; not presumed resolved; session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:proof_obligations, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f:proof_obligations	Empty arrangement and small-value checks: established by reasoning steps 56 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.
Author-listed; no exact frozen-record match; No exact frozen-record match	Reconcile the achieved/no-remaining-obligations assessments with the partial, incomplete, unresolved, and stop_no_progress fields. This is a record-consistency obligation, not an identified counterexample to the theorem. [C13]
Author-listed; no exact frozen-record match; No exact frozen-record match	Recover the complete obligation register, including the three entries not displayed in the session summary, and determine why five obligations are counted open while other extracts describe all five as established. Do not infer the missing session entries solely from artifact summaries. [C13, C14]
Author-listed; no exact frozen-record match; No exact frozen-record match	If independent verification is required, obtain and review the underlying originals or commission an explicit independent proof check. Neither activity is documented as completed in the supplied evidence. [C12, C14]
Author-listed; no exact frozen-record match; No exact frozen-record match	No specific unresolved mathematical lemma is identified by the supplied detailed deductions. Preserve that narrower observation without treating it as a resolution of the conflicting global completion statuses. [C3, C4, C7, C9, C13]

14 Conclusion

PROVISIONAL based only on covered evidence; omitted material was not reviewed: The supplied deductive argument supports M(n)=1+n(n+1)/2 for every nonnegative integer n: every arrangement satisfies the upper bound, and the first n lines y=ix+i^2 attain it. [C7, C9, C10] PROVISIONAL based only on covered evidence; omitted material was not reviewed: The main explanatory contribution is the exact insertion lemma. Convexity turns pieces of a new line into distinct split regions, making the passage from geometric structure to the triangular sum explicit

and valid even for degenerate arrangements. [C3, C4, C5, C6, C7] PROVISIONAL based only on covered evidence; omitted material was not reviewed: The empty arrangement and the values 1,2,4,7,11 provide boundary and arithmetic checks, not an empirical foundation for the theorem. The record supplies a mathematical argument but not independent verification, and its conflicting workflow statuses remain unresolved. [C11, C12, C13, C14] *Claims discussed:* C3, C4, C5, C6, C7, C9, C10, C11, C12, C13, C14

Claim status and evidence

14.1 C1 – Unresolved

The domain is n full straight lines in the Euclidean plane, with n a nonnegative integer; regions are the connected components of the complement of their union, and the empty arrangement has one region. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. These are the recorded domain and region definitions. With no lines removed, the entire plane remains a single connected component. The upper-bound argument explicitly permits repeated lines, so the presentation need not impose distinctness as an additional hypothesis. *Evidence:* project:01af8ffa-60c3-4eea-bc15-36dca347ddfa, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.2 C2 – Unresolved

The recorded proof strategy separates a universal incremental upper bound from an attaining construction; small cases serve only as checks. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The current strategy and session state this separation explicitly, and the removed draft records the same formula and construction. The draft is corroborating evidence for the argument's architecture, not evidence of a separate successful run. *Evidence:* strategy:ee7d3b31-9c8d-4b44-a1a8-b123c67e59df, strategy:16d591eb-fd1d-4719-93bc-f2e038a43f85, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa

14.3 C3 – Unresolved

Every region of a finite line arrangement is a nonempty strict-sign cell of affine line equations and is therefore open and convex. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The recorded identification uses the two strict half-planes determined by each line. A connected component cannot change the sign of a line equation without meeting that line. Conversely, a nonempty intersection of the chosen open half-planes is convex, hence connected, and lies in the complement. Thus these nonempty cells are precisely the regions, rather than merely subsets of them. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.4 C4 – Unresolved

A noncoincident added line meeting the existing lines at m distinct points creates exactly m+1 new regions. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The m points divide the new line into m+1 connected pieces, each contained in an old region. Two pieces cannot belong to the same old region: convexity would force the segment between their points, including an intervening old-line intersection, to remain in that region. Each encountered region is open, so a point on the new line has nearby region points on both sides. Its intersections with the two strict half-planes are nonempty and convex, giving exactly two replacement regions. Each encountered old region therefore contributes one net new region. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.5 C5 – Unresolved

A line coincident with an existing line adds zero regions; parallelism and concurrency do not invalidate the distinct-intersection counting argument. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. A coincident insertion leaves the removed union unchanged and must be handled separately from the noncoincident increment lemma. Parallel predecessors contribute no intersection point, and concurrent intersections count as one distinct point rather than several. These cases can only lower the available intersection count. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa

14.6 C6 – Unresolved

In any insertion order, the kth line adds at most k regions. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. If the kth line is noncoincident, each of its k-1 predecessors contributes at most one intersection point, so m $\leq$ k-1. The increment lemma gives m+1 $\leq$ k. If it is coincident, its zero increment also satisfies the bound. No general-position assumption is used. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.7 C7 – Unresolved

Every arrangement of n lines has at most 1+n(n+1)/2 regions. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. Let A_k be the region count after the first k lines in an arbitrary arrangement. Starting from A_0=1, the recorded increment bound gives A_k-A_{k-1} $\leq$ k. Summing these inequalities yields A_n $\leq$ 1+_{k=1}^n k = 1+n(n+1)/2. At n=0 the sum is empty and gives one. This is an upper bound, not yet an assertion that every increment can be attained. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, strategy:16d591eb-fd1d-4719-93bc-f2e038a43f85

14.8 C8 – Unresolved

For the family L_i: y=ix+i^2, distinct slopes exclude parallel or coincident pairs, and L_i intersects L_k at (-(i+k),-ik) when i<k. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. Equating ix+i^2 and kx+k^2 gives (i-k)x=(k-i)(k+i), hence x=-(i+k). Substitution gives y=-ik. These are the intersection coordinates supplied in the saved deductions. The differing slopes ensure that each such pair has a unique intersection. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa

14.9 C9 – Unresolved

The first n lines of L_i: y=ix+i^2 attain 1+n(n+1)/2 regions for every n≥0. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. For fixed k, different predecessors i have different x-coordinates -(i+k), so L_k has exactly k-1 distinct old-line intersections. The increment lemma therefore gives exactly k new regions. This includes k=1, with no intersections and one new region. Starting from the empty arrangement and summing these exact increments proves attainment for every finite prefix. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.10 C10 – Unresolved

The maximum number of regions is M(n)=1+n(n+1)/2 for every nonnegative integer n. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The universal upper bound in C7 applies to every allowable arrangement, while C9 gives an arrangement reaching that same value for every n. Together they establish the extremal formula. The established label describes the supplied deductive support; it does not assert independent or machine-checked verification. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa

14.11 C11 – Unresolved

The recorded small-value checks are M(0),,M(4)=1,2,4,7,11, and they are not used to prove the general formula. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The values agree with the initial count one and successive increments 1,2,3,4. The evidence explicitly classifies these as deductive checks rather than a basis for extrapolation or a computational experiment. *Evidence:* artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa

14.12 C12 – Limitation

The supplied record contains no computational evidence or external formal verification; Scientific Python and SageMath were recorded as unavailable. The extracts report null code or computational feedback, empty computational citations, and no persisted executions. The reviewer calls

the argument convincing but unverified. These facts limit claims about validation provenance; they do not themselves contradict the mathematical deductions. *Evidence:* session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.13 C13 – Unresolved

The record's overall completion and obligation statuses are internally inconsistent, and their reconciliation is unresolved. Some fields accept all six reasoning steps, describe the objective as achieved, and report no remaining obligations or objections. Other fields mark the outcome partial, mixed, unresolved, or incomplete, and record stop_no_progress. A session summary counts five open obligations while displayed entries call obligations established; artifact extracts report all five established. The index supplies no explanation that would justify selecting one global status as definitive. *Evidence:* session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f

14.14 C14 – Limitation

The supplied evidence index supports a source-linked synthesis, but not a claim of exhaustive review of the underlying originals or independent corroboration by multiple sources. The package identifies the extracts as untrusted summaries rather than original verification, and several entries are visibly truncated. Repeated descriptions of the same session and artifact are not independent replications. An early session extract says an artifact's contents are absent, whereas later artifact-linked extracts supply substantial mathematical content; that local absence should not be generalized to the whole package. *Evidence:* session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa, artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, strategy:16d591eb-fd1d-4719-93bc-f2e038a43f85

A Structured reasoning and review records

The following material is transcribed from the frozen research snapshot to preserve recorded reasoning, Skeptic assessments, negative findings, contradictions, and unresolved objections that may not fit in the synthesized narrative. These records are not independent verification.

A.1 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/0/provider_input_audit/request_body

{ "items": { "type": "string" }, "maxItems": 8, "title": "Basis Artifact Ids", "type": "array" }

A.2 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/0/provider_input_audit/request_body

{ "maxLength": 6000, "minLength": 1, "title": "Method", "type": "string" }

A.3 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/0/provider_input_audit/request_body

{ "items": { "$ref": "#/$defs/ResearchReasoningStep" }, "maxItems": 24, "title": "Reasoning Steps", "type": "array" }

A.4 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/0/provider_input_audit/request_body

{ "items": { "type": "string" }, "maxItems": 12, "title": "Open Obligations", "type": "array" }

A.5 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/0/output/method

"Test the candidate formula by proving a universal bound on the increase caused by each new line, then separately construct lines attaining every incremental bound. The discriminating condition for equality at step k is that the new line meets the preceding k-1 lines at k-1 distinct points. Regions are connected components of the plane after removing the full lines. The argument is deductive; small cases are checks, not evidence for the general theorem. No computation is needed or available."

A.6 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/0/output/reasoning_steps

[{ "title": "Regions of a line arrangement are convex", "argument": "For finitely many lines, choose an affine equation for each line. On any connected component of their complement, each equation has a constant strict sign, since changing sign would require crossing its zero set. Conversely, the set realizing any fixed collection of strict signs is an intersection of open half-planes, hence is convex and, if nonempty, connected. Thus the regions are exactly the nonempty such intersections. In particular, each region is open and convex. With no lines, the entire plane is one region.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "An added line creates exactly one new region per piece cut out on it", "argument": "Let L be a new line not coinciding with an existing line, and let m be the number of distinct points at which it meets the existing lines. Removing these points divides L into m+1 connected pieces, including unbounded pieces. Each piece lies in one old region, because none of the old line equations vanishes on it and their signs are constant along it. Different pieces lie in different old regions: otherwise convexity would put the segment joining points from those pieces inside that region, although that segment contains an old-line intersection point. Each region met by a piece is split into exactly two regions. Indeed, an interior point of the piece has a small disk inside the old region, so the region has points on both sides of L; its intersections with the two open half-planes bounded by L are nonempty and convex, and no path between them can avoid L. Old regions not met by L remain unchanged. Therefore adding L increases the region count by exactly m+1. A line coinciding with an existing line instead increases it by zero.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "Universal upper bound", "argument": "When adding the kth line, each of the preceding k-1 lines contributes at most one intersection point unless it coincides with the new line. For a noncoincident new line, m $\leq$ k-1, so the preceding lemma bounds the increase by k. A coincident line also satisfies this bound because its increase is zero. Starting from one region, every arrangement of n lines therefore has at most 1 + (k=1 to n) k = 1 + n(n+1)/2 regions. Parallel lines and multiple lines through one point cannot invalidate this bound: they only reduce the number of distinct intersection points on an added line.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "An explicit

construction reaches every incremental bound", "argument": "For each positive integer i, define L_i by y = ix + i^2. These lines have distinct slopes, so no two are parallel or coincident. For i < k, solving the equations of L_i and L_k gives x = -(i+k) and y = -ik. With k fixed, the x-coordinates -(i+k), for i = 1, , k-1, are all distinct. Hence L_k meets the preceding k-1 lines at exactly k-1 distinct points. By the increment lemma, adding L_k creates exactly k new regions, including k=1, where there are no intersection points and the first line creates one new region. This construction works for every finite n by taking its first n lines.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "Matching lower bound and the empty arrangement", "argument": "The explicit construction has exactly 1 + (k=1 to n) k = 1 + n(n+1)/2 regions, giving a lower bound equal to the universal upper bound. Thus this value is the maximum. For n=0 the construction is empty, the sum is empty, and the plane has exactly one region, agreeing with the formula.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "Small-value checks", "argument": "For n = 0, 1, 2, 3, 4, the formula gives 1, 2, 4, 7, 11 regions. These agree respectively with the empty plane, one dividing line, two intersecting lines, three pairwise-intersecting nonconcurrent lines, and a fourth line crossing the first three at distinct points. In the explicit construction the successive increases are 1, 2, 3, 4, confirming these checks without using them to infer the general result.", "epistemic_status": "deduction", "basis_artifact_ids": [] }]

A.7 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/provider_input_audit/request_body

{ "items": { "type": "string" }, "maxItems": 12, "title": "Objections", "type": "array" }

A.8 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/provider_input_audit/request_body

{ "items": { "type": "string" }, "maxItems": 12, "title": "Remaining Obligations", "type": "array" }

A.9 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/provider_input_audit/request_body

{ "items": { "$ref": "#/$defs/ReasoningAssessment" }, "maxItems": 24, "title": "Reasoning Assessments", "type": "array" }

A.10 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/provider_input_audit/request_bod

{ "items": { "type": "string" }, "maxItems": 12, "title": "Proof Obligations", "type": "array" }

A.11 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/provider_input_audit/request_bod

{ "items": { "type": "string" }, "maxItems": 12, "title": "Unresolved Objections", "type": "array" }

A.12 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/provider_input_audit/request_bod

{ "pattern": "^(convincing_unverified|incomplete)$", "title": "Verification Status", "type": "string" }

A.13 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/output/reasoning_assessments

[{ "step_number": 1, "disposition": "accepted", "rationale": "A continuous affine equation cannot change sign on a connected subset avoiding its zero line. Conversely, every nonempty strict-sign cell is an intersection of open half-planes and hence is open and convex. These two facts identify the cells with the connected components, including the whole plane when there are no lines.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 2, "disposition": "accepted", "rationale": "Removing m distinct intersection points from a full line leaves m+1 intervals or rays. Each lies in an old sign cell. Two different pieces cannot lie in the same convex old region because the joining segment would contain a removed intersection point. Any old region meeting the new line contains a disk around such a meeting point, so both open half-plane intersections are nonempty; each is convex, and together they are exactly its two new components. Coincident lines leave the union unchanged.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 3, "disposition": "accepted", "rationale": "A line not coinciding with an earlier line meets each earlier line in at most one point, so m ≤ k-1. The increment lemma gives an increase of at most k; a coincident line contributes zero. Summing these bounds from the initial single region yields 1 + n(n+1)/2 without imposing general position.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 4, "disposition": "accepted", "rationale": "Distinct integer slopes ensure no parallel or coincident pairs. Solving ix+i^2 = kx+k^2 gives x = -(i+k) and y = -ik. For fixed k the x-coordinates are distinct for distinct preceding indices i. Thus the kth line has exactly k-1 distinct old-line intersection points and adds exactly k regions, including the first-line case.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 5, "disposition": "accepted", "rationale": "The construction realizes every increment used in the upper bound, so its region count equals that bound for every finite n. At n = 0 the empty union leaves the plane connected and the empty sum is zero, giving one region.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 6, "disposition": "accepted", "rationale": "Direct substitution gives 1, 2, 4, 7, 11 for n = 0, 1, 2, 3, 4. The stated geometric configurations realize the corresponding increments, and these checks are correctly kept separate from the general proof.", "objection": "", "basis_artifact_ids": [] }]

A.14 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/output/proof_obligations

["Convexity and identification of arrangement regions: established by reasoning step 1 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Exact increment m+1 for a noncoincident line and zero for a coincident line: established by reasoning step 2 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Universal upper bound for arbitrary arrangements: established by reasoning step 3 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Existence of an attaining arrangement for every positive integer n: established by the intersection calculation and increment argument in reasoning steps 45 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Empty arrangement and small-value checks: established by reasoning steps 56 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1."]

A.15 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: steps/1/output/verification_status

"convincing_unverified"

A.16 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: strategy_assessments/0/basis_artifact_ids

["4b50ca5d-ad1f-4f6a-8042-81975427938f"]

A.17 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: outcome

{ "assessment": "mixed", "label": "Mixed · partial progress, unresolved", "summary": "For every integer n ≥ 0, the maximum number of regions formed by n full straight lines in the Euclidean plane is 1 + n(n+1)/2, attained by the lines y = ix + i^2 for i = 1, , n.", "caveats": ["Agent-reported evidence is unverified; agreement is not independent verification.", "The saved argument remains incomplete; no solution is claimed."], "basis_artifact_ids": ["4b50ca5d-ad1f-4f6a-8042-81975427938f"], "provisional": false, "solution_review": { "status": "partial", "claim": "For every integer n ≥ 0, the maximum number of regions formed by n full straight lines in the Euclidean plane is 1 + n(n+1)/2, attained by the lines y = ix + i^2 for i = 1, , n.", "claim_scope": "unconditional", "argument_summary": "Cycle 1, Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, supplies the complete argument. Regions are convex sign cells. A noncoincident added line with m distinct old-line intersection points passes through m+1 different old regions and splits each into two, increasing the count by m+1. Consequently, the kth line adds at most k regions. Summing gives the upper bound. For y = ix + i^2, the intersection of lines i and k has coordinates (-(i+k), -ik); these intersections are distinct along line k as i varies. Every increment therefore reaches k, establishing equality. The empty arrangement gives one region, and the checked values for n = 0 through 4 are 1, 2, 4, 7, 11.", "argument_steps": [{ "number": 1, "title": "Regions of a line arrangement are convex", "argument": "For finitely many lines, choose an affine equation for each line. On any connected component of their complement, each equation has a constant strict sign, since changing sign would require crossing its zero set. Conversely, the set realizing any fixed collection of strict signs is an intersection of open half-planes, hence is convex and, if nonempty, connected. Thus the regions are exactly the nonempty such intersections. In particular, each region is open and convex. With no lines, the entire plane is one region.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A continuous affine equation cannot change sign on a connected subset avoiding its zero line. Conversely, every nonempty strict-sign cell is an intersection of open half-planes and hence is open and convex. These two facts identify the cells with the connected components, including the whole plane when there are no lines.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 2, "title": "An added line creates exactly one new region per piece cut out on it", "argument": "Let L be a new line not coinciding with an existing line, and let m be the number of distinct points at which it meets the existing lines. Removing these points divides L into m+1 connected pieces, including unbounded pieces. Each piece lies in one old region, because none of the old line equations vanishes on it and their signs are constant along it. Different pieces lie in different old regions: otherwise convexity would put the segment joining points from those pieces inside that region, although that segment contains an old-line intersection point. Each region met by a piece is split into exactly two regions. Indeed, an interior point of the piece has a small disk inside the old region, so the region has points on both sides of L; its intersections with the two open half-planes bounded by L are nonempty and convex, and no path between them can avoid L. Old regions not met by L remain unchanged. Therefore adding L increases the region count by exactly m+1. A line coinciding with an existing line instead increases it by zero.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Removing m distinct intersection points from a full line leaves m+1 intervals or rays. Each lies in an old sign cell. Two

different pieces cannot lie in the same convex old region because the joining segment would contain a removed intersection point. Any old region meeting the new line contains a disk around such a meeting point, so both open half-plane intersections are nonempty; each is convex, and together they are exactly its two new components. Coincident lines leave the union unchanged.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 3, "title": "Universal upper bound", "argument": "When adding the kth line, each of the preceding k-1 lines contributes at most one intersection point unless it coincides with the new line. For a noncoincident new line, m $\leq$ k-1, so the preceding lemma bounds the increase by k. A coincident line also satisfies this bound because its increase is zero. Starting from one region, every arrangement of n lines therefore has at most 1 + (k=1 to n) k = 1 + n(n+1)/2 regions. Parallel lines and multiple lines through one point cannot invalidate this bound: they only reduce the number of distinct intersection points on an added line.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A line not coinciding with an earlier line meets each earlier line in at most one point, so m $\leq$ k-1. The increment lemma gives an increase of at most k; a coincident line contributes zero. Summing these bounds from the initial single region yields 1 + n(n+1)/2 without imposing general position.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 4, "title": "An explicit construction reaches every incremental bound", "argument": "For each positive integer i, define L_i by y = ix + i^2. These lines have distinct slopes, so no two are parallel or coincident. For i < k, solving the equations of L_i and L_k gives x = -(i+k) and y = -ik. With k fixed, the x-coordinates -(i+k), for i = 1, , k-1, are all distinct. Hence L_k meets the preceding k-1 lines at exactly k-1 distinct points. By the increment lemma, adding L_k creates exactly k new regions, including k=1, where there are no intersection points and the first line creates one new region. This construction works for every finite n by taking its first n lines.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Distinct integer slopes ensure no parallel or coincident pairs. Solving ix+i^2 = kx+k^2 gives x = -(i+k) and y = -ik. For fixed k the x-coordinates are distinct for distinct preceding indices i. Thus the kth line has exactly k-1 distinct old-line intersection points and adds exactly k regions, including the first-line case.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 5, "title": "Matching lower bound and the empty arrangement", "argument": "The explicit construction has exactly 1 + (k=1 to n) k = 1 + n(n+1)/2 regions, giving a lower bound equal to the universal upper bound. Thus this value is the maximum. For n=0 the construction is empty, the sum is empty, and the plane has exactly one region, agreeing with the formula.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "The construction realizes every increment used in the upper bound, so its region count equals that bound for every finite n. At n = 0 the empty union leaves the plane connected and the empty sum is zero, giving one region.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 6, "title": "Small-value checks", "argument": "For n = 0, 1, 2, 3, 4, the formula gives 1, 2, 4, 7, 11 regions. These agree respectively with the empty plane, one dividing line, two intersecting lines, three pairwise-intersecting nonconcurrent lines, and a fourth line crossing the first three at distinct points. In the explicit construction the successive increases are 1, 2, 3, 4, confirming these checks without using them to infer the general result.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Direct substitution gives 1, 2, 4, 7, 11 for n = 0, 1, 2, 3, 4. The stated geometric configurations realize the corresponding increments, and these checks are correctly kept separate from the general proof.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }], "assumptions": ["n is a nonnegative integer and the objects are full straight lines in the Euclidean

plane.", "Regions are connected components of the complement of the union of the lines.", "Parallel, concurrent, and repeated lines are allowed in the universal upper-bound argument; the construction uses distinct lines.", "These are the stated problem domain and definitions, not additional unproved geometric hypotheses."], "proof_obligations": ["Convexity and identification of arrangement regions: established by reasoning step 1 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Exact increment m+1 for a noncoincident line and zero for a coincident line: established by reasoning step 2 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Universal upper bound for arbitrary arrangements: established by reasoning step 3 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Existence of an attaining arrangement for every positive integer n: established by the intersection calculation and increment argument in reasoning steps 45 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Empty arrangement and small-value checks: established by reasoning steps 56 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1."], "verification_status": "incomplete", "unresolved_objections": [], "resolved_objections": [], "basis_artifact_ids": ["4b50ca5d-ad1f-4f6a-8042-81975427938f"], "skeptic_response": "The maximum is M(n) = 1 + n(n+1)/2 for every nonnegative integer n. The supplied proof establishes both the universal upper bound and an attaining construction. Its key increment lemma correctly handles parallel lines, multiple intersections, and repeated lines. The explicit family y = ix + i^2 attains the bound because its kth line meets the preceding lines at k-1 distinct points. The empty arrangement and small values are also covered. No material objection remains; this is a convincing deductive argument, not an independently verified proof.", "skeptic_objections": [], "open_obligations": [], "strategy_findings": [{ "strategy_id": "ee7d3b31-9c8d-4b44-a1a8-b123c67e59df", "strategy_title": "Bound the new regions, then seek equality", "status": "partial", "execution_status": "ended", "failure_reason": "Stop no progress", "claim": "For every integer n $\geq$ 0, the maximum number of regions formed by n full straight lines in the Euclidean plane is 1 + n(n+1)/2, attained by the lines y = ix + i^2 for i = 1, , n.", "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f", "research_artifact_ids": ["4b50ca5d-ad1f-4f6a-8042-81975427938f"], "reasoning_steps": [{ "number": 1, "title": "Regions of a line arrangement are convex", "argument": "For finitely many lines, choose an affine equation for each line. On any connected component of their complement, each equation has a constant strict sign, since changing sign would require crossing its zero set. Conversely, the set realizing any fixed collection of strict signs is an intersection of open half-planes, hence is convex and, if nonempty, connected. Thus the regions are exactly the nonempty such intersections. In particular, each region is open and convex. With no lines, the entire plane is one region.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A continuous affine equation cannot change sign on a connected subset avoiding its zero line. Conversely, every nonempty strict-sign cell is an intersection of open half-planes and hence is open and convex. These two facts identify the cells with the connected components, including the whole plane when there are no lines.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 2, "title": "An added line creates exactly one new region per piece cut out on it", "argument": "Let L be a new line not coinciding with an existing line, and let m be the number of distinct points at which it meets the existing lines. Removing these points divides L into m+1 connected pieces, including unbounded pieces. Each piece lies in one old region, because none of the old line equations vanishes on it and their signs are constant along it. Different pieces lie in different old regions: otherwise convexity would put the segment joining points from those pieces inside that region, although that segment contains an old-line intersection point. Each region met by a piece is split into exactly two regions. Indeed, an interior point of the piece has a small disk inside the old region, so the region has points on both sides of L; its intersections with the two open half-planes bounded by L are nonempty and convex, and no path between them can avoid L. Old regions not

met by L remain unchanged. Therefore adding L increases the region count by exactly m+1. A line coinciding with an existing line instead increases it by zero.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Removing m distinct intersection points from a full line leaves m+1 intervals or rays. Each lies in an old sign cell. Two different pieces cannot lie in the same convex old region because the joining segment would contain a removed intersection point. Any old region meeting the new line contains a disk around such a meeting point, so both open half-plane intersections are nonempty; each is convex, and together they are exactly its two new components. Coincident lines leave the union unchanged.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 3, "title": "Universal upper bound", "argument": "When adding the kth line, each of the preceding k-1 lines contributes at most one intersection point unless it coincides with the new line. For a noncoincident new line, m $\leq$ k-1, so the preceding lemma bounds the increase by k. A coincident line also satisfies this bound because its increase is zero. Starting from one region, every arrangement of n lines therefore has at most 1 + (k=1 to n) k = 1 + n(n+1)/2 regions. Parallel lines and multiple lines through one point cannot invalidate this bound: they only reduce the number of distinct intersection points on an added line.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A line not coinciding with an earlier line meets each earlier line in at most one point, so m $\leq$ k-1. The increment lemma gives an increase of at most k; a coincident line contributes zero. Summing these bounds from the initial single region yields 1 + n(n+1)/2 without imposing general position.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 4, "title": "An explicit construction reaches every incremental bound", "argument": "For each positive integer i, define L_i by y = ix + i^2. These lines have distinct slopes, so no two are parallel or coincident. For i < k, solving the equations of L_i and L_k gives x = -(i+k) and y = -ik. With k fixed, the x-coordinates -(i+k), for i = 1, , k-1, are all distinct. Hence L_k meets the preceding k-1 lines at exactly k-1 distinct points. By the increment lemma, adding L_k creates exactly k new regions, including k=1, where there are no intersection points and the first line creates one new region. This construction works for every finite n by taking its first n lines.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Distinct integer slopes ensure no parallel or coincident pairs. Solving ix+i^2 = kx+k^2 gives x = -(i+k) and y = -ik. For fixed k the x-coordinates are distinct for distinct preceding indices i. Thus the kth line has exactly k-1 distinct old-line intersection points and adds exactly k regions, including the first-line case.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 5, "title": "Matching lower bound and the empty arrangement", "argument": "The explicit construction has exactly 1 + (k=1 to n) k = 1 + n(n+1)/2 regions, giving a lower bound equal to the universal upper bound. Thus this value is the maximum. For n=0 the construction is empty, the sum is empty, and the plane has exactly one region, agreeing with the formula.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "The construction realizes every increment used in the upper bound, so its region count equals that bound for every finite n. At n = 0 the empty union leaves the plane connected and the empty sum is zero, giving one region.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 6, "title": "Small-value checks", "argument": "For n = 0, 1, 2, 3, 4, the formula gives 1, 2, 4, 7, 11 regions. These agree respectively with the empty plane, one dividing line, two intersecting lines, three pairwise-intersecting nonconcurrent lines, and a fourth line crossing the first three at distinct points. In the explicit construction the successive increases are 1, 2, 3, 4, confirming these checks without using them to infer the general result.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Direct substitution gives 1, 2, 4, 7, 11 for n =

0, 1, 2, 3, 4. The stated geometric configurations realize the corresponding increments, and these checks are correctly kept separate from the general proof.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }], "skeptic_response": "The maximum is M(n) = 1 + n(n+1)/2 for every nonnegative integer n. The supplied proof establishes both the universal upper bound and an attaining construction. Its key increment lemma correctly handles parallel lines, multiple intersections, and repeated lines. The explicit family y = ix + i^2 attains the bound because its kth line meets the preceding lines at k-1 distinct points. The empty arrangement and small values are also covered. No material objection remains; this is a convincing deductive argument, not an independently verified proof.", "objections": [], "open_obligations": [] }] }, "conclusion": "### Partial finding\n\nFor every integer n ≥ 0, the maximum number of regions formed by n full straight lines in the Euclidean plane is 1 + n(n+1)/2, attained by the lines y = ix + i^2 for i = 1, , n.\n\nCycle 1, Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, supplies the complete argument. Regions are convex sign cells. A noncoincident added line with m distinct old-line intersection points passes through m+1 different old regions and splits each into two, increasing the count by m+1. Consequently, the kth line adds at most k regions. Summing gives the upper bound. For y = ix + i^2, the intersection of lines i and k has coordinates (-(i+k), -ik); these intersections are distinct along line k as i varies. Every increment therefore reaches k, establishing equality. The empty arrangement gives one region, and the checked values for n = 0 through 4 are 1, 2, 4, 7, 11.\n\nThe saved review does not establish a complete answer to the project question.\n\n### Session caveats\n- Agent-reported evidence is unverified; agreement is not independent verification.\n- The saved argument remains incomplete; no solution is claimed.\n\n**Evidence:** Artifact 1\n\n### Verification and scope\n\nVerification: Incomplete.\n\nThis Conclusion summarizes agent-reported findings for this session. SIGMA has not independently verified the result; inspect the detailed Outcome and saved evidence before relying on it.", "verification_label": "Incomplete", "argument_availability": { "status": "structured", "label": "Structured argument steps are recorded; review their evidence and objections." }, "execution_provenance": { "records": [], "unidentified_count": 0, "completed_count": 0, "completed_without_recorded_io_count": 0, "other_count": 0, "note": "No persisted Python/SageMath execution records; authored measurement claims are not established by a saved computation." }, "metrics": { "operational": [{ "label": "Backend build", "value": "fbe7e1283a6d7f9f6aaf0708342fae2502cde378", "detail": "The backend commit is captured at creation and at each provider attempt. Older sessions are never retroactively assigned the current checkout version." }, { "label": "Participating agents", "value": "Investigator, Skeptic" }, { "label": "Model calls by status", "value": "2 dispatched · 2 completed · 0 rejected · 0 timed out · 0 interrupted / cancelled · 0 failed · 0 in progress · 0 not dispatched · 0 status unavailable" }, { "label": "Tokens · input / output / cached / total", "value": "10,200 / 3,456 / 0 / 13,656", "detail": "Cached tokens are a subset of input tokens." }, { "label": "Wall-clock / agent call time", "value": "1m 16s wall · Investigator 38s across 1 call(s) · Skeptic 38s across 1 call(s)", "detail": "Agent durations sum distinct call intervals and may overlap under concurrency." }, { "label": "SageMath runs by status", "value": "0 executions", "detail": "Distinct Sage execution IDs; recovered/reused results count once. Exact computation is not proof verification." }, { "label": "Python runs by status", "value": "0 executions", "detail": "Unique persisted execution IDs; recovered or reused results count once." }, { "label": "Artifacts and storage", "value": "1 finding(s) · 0 Python file(s) (0 B) · session record 103,366 B", "detail": "Captured file sizes are summed once per execution ID and file hash/name." }, { "label": "Provider failures / retries", "value": "0 provider failure(s) (0 rejected, 0 timed out, 0 other) · 0 interrupted / cancelled · 0 later attempt(s); no automatic retries" }, { "label": "Estimated monetary cost", "value": " (plan allowance consumption unknown)" }], "research_quality": [{ "label": "Strategies investigated", "value": "Bound the new regions, then seek equality" }, { "label": "Research cycles", "value": "1

completed / 1 started", "detail": "Counts a strategy's cycle once even when its Investigator and Skeptic calls overlap with other strategies." }, { "label": "Outcome classification", "value": "Mixed · partial progress, unresolved" }, { "label": "Verification status", "value": "Incomplete", "detail": "Engineering completion, agent agreement, and successful computation do not establish mathematical verification." }, { "label": "Open proof obligations", "value": "5 · Convexity and identification of arrangement regions: established by reasoning step 1 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.; Exact increment m+1 for a noncoincident line and zero for a coincident line: established by reasoning step 2 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1. (+3 more)" }, { "label": "Evidence-backed conclusions", "value": "1 conclusion(s) · 1 cited finding(s)" }, { "label": "Strategy contribution", "value": "productive: 1" }, { "label": "Unresolved objections", "value": "0 · None recorded" }], "sources": { "session_id": "f60b98af-2fd0-493d-8eb9-1b7b204be8fa", "step_ids": ["5ae10b4f-29aa-4770-91a8-aa7d378719e3", "75aaa1f1-b3c0-4315-90b0-58ab81826e26"], "artifact_ids": ["4b50ca5d-ad1f-4f6a-8042-81975427938f"], "execution_ids": [], "strategy_ids": ["ee7d3b31-9c8d-4b44-a1a8-b123c67e59df"] } } }

A.18 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: outcome/caveats

["Agent-reported evidence is unverified; agreement is not independent verification.", "The saved argument remains incomplete; no solution is claimed."]

A.19 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: outcome/solution_review/argument_steps

[{ "number": 1, "title": "Regions of a line arrangement are convex", "argument": "For finitely many lines, choose an affine equation for each line. On any connected component of their complement, each equation has a constant strict sign, since changing sign would require crossing its zero set. Conversely, the set realizing any fixed collection of strict signs is an intersection of open half-planes, hence is convex and, if nonempty, connected. Thus the regions are exactly the nonempty such intersections. In particular, each region is open and convex. With no lines, the entire plane is one region.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A continuous affine equation cannot change sign on a connected subset avoiding its zero line. Conversely, every nonempty strict-sign cell is an intersection of open half-planes and hence is open and convex. These two facts identify the cells with the connected components, including the whole plane when there are no lines.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 2, "title": "An added line creates exactly one new region per piece cut out on it", "argument": "Let L be a new line not coinciding with an existing line, and let m be the number of distinct points at which it meets the existing lines. Removing these points divides L into m+1 connected pieces, including unbounded pieces. Each piece lies in one old region, because none of the old line equations vanishes on it and their signs are constant along it. Different pieces lie in different old regions: otherwise convexity would put the segment joining points from those pieces inside that region, although that segment contains an old-line intersection point. Each region met by a piece is split into exactly two regions. Indeed, an interior point of the piece has a small disk inside the old region, so the region has points on both sides of L; its intersections with the two open half-planes bounded by L are nonempty and convex, and no path between them can avoid L. Old regions not met by L remain unchanged. Therefore adding L increases the region count by exactly m+1. A line coinciding with an existing line instead increases it by zero.", "epistemic_status": "deduction", "basis_artifact_ids": [], "re-

view": { "disposition": "accepted", "rationale": "Removing m distinct intersection points from a full line leaves m+1 intervals or rays. Each lies in an old sign cell. Two different pieces cannot lie in the same convex old region because the joining segment would contain a removed intersection point. Any old region meeting the new line contains a disk around such a meeting point, so both open half-plane intersections are nonempty; each is convex, and together they are exactly its two new components. Coincident lines leave the union unchanged.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 3, "title": "Universal upper bound", "argument": "When adding the kth line, each of the preceding k-1 lines contributes at most one intersection point unless it coincides with the new line. For a noncoincident new line, m $\leq$ k-1, so the preceding lemma bounds the increase by k. A coincident line also satisfies this bound because its increase is zero. Starting from one region, every arrangement of n lines therefore has at most 1 + (k=1 to n) k = 1 + n(n+1)/2 regions. Parallel lines and multiple lines through one point cannot invalidate this bound: they only reduce the number of distinct intersection points on an added line.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A line not coinciding with an earlier line meets each earlier line in at most one point, so m $\leq$ k-1. The increment lemma gives an increase of at most k; a coincident line contributes zero. Summing these bounds from the initial single region yields 1 + n(n+1)/2 without imposing general position.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 4, "title": "An explicit construction reaches every incremental bound", "argument": "For each positive integer i, define L_i by y = ix + i^2. These lines have distinct slopes, so no two are parallel or coincident. For i < k, solving the equations of L_i and L_k gives x = -(i+k) and y = -ik. With k fixed, the x-coordinates -(i+k), for i = 1, , k-1, are all distinct. Hence L_k meets the preceding k-1 lines at exactly k-1 distinct points. By the increment lemma, adding L_k creates exactly k new regions, including k=1, where there are no intersection points and the first line creates one new region. This construction works for every finite n by taking its first n lines.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Distinct integer slopes ensure no parallel or coincident pairs. Solving ix+i^2 = kx+k^2 gives x = -(i+k) and y = -ik. For fixed k the x-coordinates are distinct for distinct preceding indices i. Thus the kth line has exactly k-1 distinct old-line intersection points and adds exactly k regions, including the first-line case.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 5, "title": "Matching lower bound and the empty arrangement", "argument": "The explicit construction has exactly 1 + (k=1 to n) k = 1 + n(n+1)/2 regions, giving a lower bound equal to the universal upper bound. Thus this value is the maximum. For n=0 the construction is empty, the sum is empty, and the plane has exactly one region, agreeing with the formula.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "The construction realizes every increment used in the upper bound, so its region count equals that bound for every finite n. At n = 0 the empty union leaves the plane connected and the empty sum is zero, giving one region.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 6, "title": "Small-value checks", "argument": "For n = 0, 1, 2, 3, 4, the formula gives 1, 2, 4, 7, 11 regions. These agree respectively with the empty plane, one dividing line, two intersecting lines, three pairwise-intersecting nonconcurrent lines, and a fourth line crossing the first three at distinct points. In the explicit construction the successive increases are 1, 2, 3, 4, confirming these checks without using them to infer the general result.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Direct substitution gives 1, 2, 4, 7, 11 for n = 0, 1, 2, 3, 4. The stated geometric configurations realize the corresponding increments, and these checks are correctly kept separate from the general proof.", "objection": "", "basis_artifact_ids": []

}, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }]

A.20 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: outcome/solution_review/verification_statu

"incomplete"

A.21 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: outcome/solution_review/skeptic_response

"The maximum is M(n) = 1 + n(n+1)/2 for every nonnegative integer n. The supplied proof establishes both the universal upper bound and an attaining construction. Its key increment lemma correctly handles parallel lines, multiple intersections, and repeated lines. The explicit family y = ix + i^2 attains the bound because its kth line meets the preceding lines at k-1 distinct points. The empty arrangement and small values are also covered. No material objection remains; this is a convincing deductive argument, not an independently verified proof."

A.22 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: outcome/conclusion

"### Partial finding\n\nFor every integer n $\geq$ 0, the maximum number of regions formed by n full straight lines in the Euclidean plane is 1 + n(n+1)/2, attained by the lines y = ix + i^2 for i = 1, , n.\n\nCycle 1, Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, supplies the complete argument. Regions are convex sign cells. A noncoincident added line with m distinct old-line intersection points passes through m+1 different old regions and splits each into two, increasing the count by m+1. Consequently, the kth line adds at most k regions. Summing gives the upper bound. For y = ix + i^2, the intersection of lines i and k has coordinates (-(i+k), -ik); these intersections are distinct along line k as i varies. Every increment therefore reaches k, establishing equality. The empty arrangement gives one region, and the checked values for n = 0 through 4 are 1, 2, 4, 7, 11.\n\nThe saved review does not establish a complete answer to the project question.\n\n### Session caveats\n- Agent-reported evidence is unverified; agreement is not independent verification.\n- The saved argument remains incomplete; no solution is claimed.\n\n**Evidence:** Artifact 1\n\n### Verification and scope\n\nVerification: Incomplete.\n\nThis Conclusion summarizes agent-reported findings for this session. SIGMA has not independently verified the result; inspect the detailed Outcome and saved evidence before relying on it."

A.23 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: portfolio_decision/limitations

["Cross-strategy relations are model-reported and require human review.", "Suggestions never edit, disable, combine, or remove saved strategies.", "A provider failure is not treated as evidence against an approach."]

A.24 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: method

"Test the candidate formula by proving a universal bound on the increase caused by each new line, then separately construct lines attaining every incremental bound. The discriminating condition

for equality at step k is that the new line meets the preceding k-1 lines at k-1 distinct points. Regions are connected components of the plane after removing the full lines. The argument is deductive; small cases are checks, not evidence for the general theorem. No computation is needed or available."

A.25 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: skeptic_response

"The maximum is M(n) = 1 + n(n+1)/2 for every nonnegative integer n. The supplied proof establishes both the universal upper bound and an attaining construction. Its key increment lemma correctly handles parallel lines, multiple intersections, and repeated lines. The explicit family y = ix + i^2 attains the bound because its kth line meets the preceding lines at k-1 distinct points. The empty arrangement and small values are also covered. No material objection remains; this is a convincing deductive argument, not an independently verified proof."

A.26 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: investigator_output/reasoning_steps

[{ "title": "Regions of a line arrangement are convex", "argument": "For finitely many lines, choose an affine equation for each line. On any connected component of their complement, each equation has a constant strict sign, since changing sign would require crossing its zero set. Conversely, the set realizing any fixed collection of strict signs is an intersection of open half-planes, hence is convex and, if nonempty, connected. Thus the regions are exactly the nonempty such intersections. In particular, each region is open and convex. With no lines, the entire plane is one region.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "An added line creates exactly one new region per piece cut out on it", "argument": "Let L be a new line not coinciding with an existing line, and let m be the number of distinct points at which it meets the existing lines. Removing these points divides L into m+1 connected pieces, including unbounded pieces. Each piece lies in one old region, because none of the old line equations vanishes on it and their signs are constant along it. Different pieces lie in different old regions: otherwise convexity would put the segment joining points from those pieces inside that region, although that segment contains an old-line intersection point. Each region met by a piece is split into exactly two regions. Indeed, an interior point of the piece has a small disk inside the old region, so the region has points on both sides of L; its intersections with the two open half-planes bounded by L are nonempty and convex, and no path between them can avoid L. Old regions not met by L remain unchanged. Therefore adding L increases the region count by exactly m+1. A line coinciding with an existing line instead increases it by zero.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "Universal upper bound", "argument": "When adding the kth line, each of the preceding k-1 lines contributes at most one intersection point unless it coincides with the new line. For a noncoincident new line, m $\leq$ k-1, so the preceding lemma bounds the increase by k. A coincident line also satisfies this bound because its increase is zero. Starting from one region, every arrangement of n lines therefore has at most 1 + (k=1 to n) k = 1 + n(n+1)/2 regions. Parallel lines and multiple lines through one point cannot invalidate this bound: they only reduce the number of distinct intersection points on an added line.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "An explicit construction reaches every incremental bound", "argument": "For each positive integer i, define L_i by y = ix + i^2. These lines have distinct slopes, so no two are parallel or coincident. For i < k,

solving the equations of L_i and L_k gives x = -(i+k) and y = -ik. With k fixed, the x-coordinates -(i+k), for i = 1, , k-1, are all distinct. Hence L_k meets the preceding k-1 lines at exactly k-1 distinct points. By the increment lemma, adding L_k creates exactly k new regions, including k=1, where there are no intersection points and the first line creates one new region. This construction works for every finite n by taking its first n lines.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "Matching lower bound and the empty arrangement", "argument": "The explicit construction has exactly 1 + (k=1 to n) k = 1 + n(n+1)/2 regions, giving a lower bound equal to the universal upper bound. Thus this value is the maximum. For n=0 the construction is empty, the sum is empty, and the plane has exactly one region, agreeing with the formula.", "epistemic_status": "deduction", "basis_artifact_ids": [] }, { "title": "Small-value checks", "argument": "For n = 0, 1, 2, 3, 4, the formula gives 1, 2, 4, 7, 11 regions. These agree respectively with the empty plane, one dividing line, two intersecting lines, three pairwise-intersecting nonconcurrent lines, and a fourth line crossing the first three at distinct points. In the explicit construction the successive increases are 1, 2, 3, 4, confirming these checks without using them to infer the general result.", "epistemic_status": "deduction", "basis_artifact_ids": [] }]

A.27 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: skeptic_output/reasoning_assessments

[{ "step_number": 1, "disposition": "accepted", "rationale": "A continuous affine equation cannot change sign on a connected subset avoiding its zero line. Conversely, every nonempty strict-sign cell is an intersection of open half-planes and hence is open and convex. These two facts identify the cells with the connected components, including the whole plane when there are no lines.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 2, "disposition": "accepted", "rationale": "Removing m distinct intersection points from a full line leaves m+1 intervals or rays. Each lies in an old sign cell. Two different pieces cannot lie in the same convex old region because the joining segment would contain a removed intersection point. Any old region meeting the new line contains a disk around such a meeting point, so both open half-plane intersections are nonempty; each is convex, and together they are exactly its two new components. Coincident lines leave the union unchanged.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 3, "disposition": "accepted", "rationale": "A line not coinciding with an earlier line meets each earlier line in at most one point, so m $\leq$ k-1. The increment lemma gives an increase of at most k; a coincident line contributes zero. Summing these bounds from the initial single region yields 1 + n(n+1)/2 without imposing general position.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 4, "disposition": "accepted", "rationale": "Distinct integer slopes ensure no parallel or coincident pairs. Solving ix+i^2 = kx+k^2 gives x = -(i+k) and y = -ik. For fixed k the x-coordinates are distinct for distinct preceding indices i. Thus the kth line has exactly k-1 distinct old-line intersection points and adds exactly k regions, including the first-line case.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 5, "disposition": "accepted", "rationale": "The construction realizes every increment used in the upper bound, so its region count equals that bound for every finite n. At n = 0 the empty union leaves the plane connected and the empty sum is zero, giving one region.", "objection": "", "basis_artifact_ids": [] }, { "step_number": 6, "disposition": "accepted", "rationale": "Direct substitution gives 1, 2, 4, 7, 11 for n = 0, 1, 2, 3, 4. The stated geometric configurations realize the corresponding increments, and these checks are correctly kept separate from the general proof.", "objection": "", "basis_artifact_ids": [] }]

A.28 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: skeptic_output/proof_obligations

["Convexity and identification of arrangement regions: established by reasoning step 1 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Exact increment m+1 for a noncoincident line and zero for a coincident line: established by reasoning step 2 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Universal upper bound for arbitrary arrangements: established by reasoning step 3 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Existence of an attaining arrangement for every positive integer n: established by the intersection calculation and increment argument in reasoning steps 45 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1.", "Empty arrangement and small-value checks: established by reasoning steps 56 of Investigator step 5ae10b4f-29aa-4770-91a8-aa7d378719e3, cycle 1."]

A.29 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: skeptic_output/verification_status

"convincing_unverified"

A.30 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: reasoning_steps

[{ "number": 1, "title": "Regions of a line arrangement are convex", "argument": "For finitely many lines, choose an affine equation for each line. On any connected component of their complement, each equation has a constant strict sign, since changing sign would require crossing its zero set. Conversely, the set realizing any fixed collection of strict signs is an intersection of open half-planes, hence is convex and, if nonempty, connected. Thus the regions are exactly the nonempty such intersections. In particular, each region is open and convex. With no lines, the entire plane is one region.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A continuous affine equation cannot change sign on a connected subset avoiding its zero line. Conversely, every nonempty strict-sign cell is an intersection of open half-planes and hence is open and convex. These two facts identify the cells with the connected components, including the whole plane when there are no lines.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 2, "title": "An added line creates exactly one new region per piece cut out on it", "argument": "Let L be a new line not coinciding with an existing line, and let m be the number of distinct points at which it meets the existing lines. Removing these points divides L into m+1 connected pieces, including unbounded pieces. Each piece lies in one old region, because none of the old line equations vanishes on it and their signs are constant along it. Different pieces lie in different old regions: otherwise convexity would put the segment joining points from those pieces inside that region, although that segment contains an old-line intersection point. Each region met by a piece is split into exactly two regions. Indeed, an interior point of the piece has a small disk inside the old region, so the region has points on both sides of L; its intersections with the two open half-planes bounded by L are nonempty and convex, and no path between them can avoid L. Old regions not met by L remain unchanged. Therefore adding L increases the region count by exactly m+1. A line coinciding with an existing line instead increases it by zero.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Removing m distinct intersection points from a full

line leaves m+1 intervals or rays. Each lies in an old sign cell. Two different pieces cannot lie in the same convex old region because the joining segment would contain a removed intersection point. Any old region meeting the new line contains a disk around such a meeting point, so both open half-plane intersections are nonempty; each is convex, and together they are exactly its two new components. Coincident lines leave the union unchanged.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 3, "title": "Universal upper bound", "argument": "When adding the kth line, each of the preceding k-1 lines contributes at most one intersection point unless it coincides with the new line. For a noncoincident new line, m $\leq$ k-1, so the preceding lemma bounds the increase by k. A coincident line also satisfies this bound because its increase is zero. Starting from one region, every arrangement of n lines therefore has at most 1 + (k=1 to n) k = 1 + n(n+1)/2 regions. Parallel lines and multiple lines through one point cannot invalidate this bound: they only reduce the number of distinct intersection points on an added line.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "A line not coinciding with an earlier line meets each earlier line in at most one point, so m $\leq$ k-1. The increment lemma gives an increase of at most k; a coincident line contributes zero. Summing these bounds from the initial single region yields 1 + n(n+1)/2 without imposing general position.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 4, "title": "An explicit construction reaches every incremental bound", "argument": "For each positive integer i, define L_i by y = ix + i^2. These lines have distinct slopes, so no two are parallel or coincident. For i < k, solving the equations of L_i and L_k gives x = -(i+k) and y = -ik. With k fixed, the x-coordinates -(i+k), for i = 1, , k-1, are all distinct. Hence L_k meets the preceding k-1 lines at exactly k-1 distinct points. By the increment lemma, adding L_k creates exactly k new regions, including k=1, where there are no intersection points and the first line creates one new region. This construction works for every finite n by taking its first n lines.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Distinct integer slopes ensure no parallel or coincident pairs. Solving ix+i^2 = kx+k^2 gives x = -(i+k) and y = -ik. For fixed k the x-coordinates are distinct for distinct preceding indices i. Thus the kth line has exactly k-1 distinct old-line intersection points and adds exactly k regions, including the first-line case.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 5, "title": "Matching lower bound and the empty arrangement", "argument": "The explicit construction has exactly 1 + (k=1 to n) k = 1 + n(n+1)/2 regions, giving a lower bound equal to the universal upper bound. Thus this value is the maximum. For n=0 the construction is empty, the sum is empty, and the plane has exactly one region, agreeing with the formula.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "The construction realizes every increment used in the upper bound, so its region count equals that bound for every finite n. At n = 0 the empty union leaves the plane connected and the empty sum is zero, giving one region.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }, { "number": 6, "title": "Small-value checks", "argument": "For n = 0, 1, 2, 3, 4, the formula gives 1, 2, 4, 7, 11 regions. These agree respectively with the empty plane, one dividing line, two intersecting lines, three pairwise-intersecting nonconcurrent lines, and a fourth line crossing the first three at distinct points. In the explicit construction the successive increases are 1, 2, 3, 4, confirming these checks without using them to infer the general result.", "epistemic_status": "deduction", "basis_artifact_ids": [], "review": { "disposition": "accepted", "rationale": "Direct substitution gives 1, 2, 4, 7, 11 for n = 0, 1, 2, 3, 4. The stated geometric configurations realize the corresponding increments, and these checks are correctly kept separate from the general proof.", "objection": "", "basis_artifact_ids": [] }, "research_artifact_id": "4b50ca5d-ad1f-4f6a-8042-81975427938f" }]

A.31 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: objective_review/verification_status

"incomplete"

A.32 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: objective_review/basis_artifact_ids

["4b50ca5d-ad1f-4f6a-8042-81975427938f"]

A.33 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / negative

"Session 1 ended after one cycle and 2 of its 12 permitted calls. Although its status/action are recorded as ended/Completed, its stopping reason is Stop no progress and its last decision is stop_no_progress. One artifact identifier is logged, but its contents are not supplied; these execution records do not establish a session-level mathematical result."

A.34 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"Scientific Python and SageMath are both marked unavailable. Their listed configurations do not establish that any computation was performed."

A.35 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"Scientific Python and SageMath were both recorded as unavailable. This batch supplies the research request and audit/schema fields, but no research response or execution result establishing a formula, an upper-bound proof, or an attaining construction."

A.36 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"The output reports a complete self-contained deductive argument, lists no open obligations, and claims neither computational evidence nor external formal verification. Python and Sage code are null and computational citations are empty. At the subsequent skeptic step, both Scientific Python and SageMath environments are marked unavailable."

A.37 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"The Investigator reports a complete self-contained deductive argument and no open obligations, but claims neither computational evidence nor external formal verification. Python and SageMath are recorded as unavailable, with no code requested. The manifest also cautions that deterministic extracts are untrusted summaries of saved originals, not original evidence or verification."

A.38 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"The response calls the argument convincing but not independently verified, states that no material objection remains, and records an empty objections list. Computational feedback is null. Nevertheless, the assessment is 'stop' with stop_reason 'unresolved'."

A.39 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"The review recorded the objective as achieved, the claim as unconditional, all six reasoning steps as accepted, and no remaining proof obligations or unresolved objections. Nevertheless, verification_status was 'convincing_unverified'; no computational citations were supplied."

A.40 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"The record is internally inconsistent about completion: its argument summary says the complete argument is supplied and all six steps are accepted, but the overall assessment is mixed/partial progress, unresolved, the solution-review status is partial, and a caveat says the argument remains incomplete and no solution is claimed. Agent-reported evidence is unverified; agreement is not independent verification. The coordinator's unverified recommendations place 'Bound the new regions, then seek',"

A.41 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"All six structured deductive steps were accepted; objection lists and open-obligation lists are empty. The reviewer calls the argument convincing but not independently verified. Nevertheless, verification is labeled incomplete, the strategy is partial and ended with 'Stop no progress', and the conclusion says the saved argument remains incomplete and claims no solution. No persisted Python/SageMath execution records exist, all recorded execution counts are zero, and authored measurement claims "

A.42 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / limitation

"The artifact describes its argument as complete and self-contained; its skeptic endorses the upper bound, attaining construction, handling of degeneracies, and empty/small cases, with no material objection recorded. These are reported deductive assessments, not independent proof verification. Small cases are checks rather than evidence for the general theorem; no computation is needed or available, and no computational evidence or external formal verification is claimed."

A.43 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / limitation

"The session records one completed/started cycle of Bound the new regions, then seek equality, one productive strategy, and one conclusion supported by one cited finding, but classifies the outcome as mixed, partial, unresolved, and incompletely verified. Five proof obligations are counted as open,

although the two displayed entries describe region convexity/identification and the increment m+1 for a noncoincident line (zero for a coincident line) as established; the other three are not shown. "

A.44 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / limitation

"The investigator describes a complete self-contained argument. The skeptic accepts all six reasoning steps and proof obligations, judges the strategy a solution and the objective achieved, and stops with no material or unresolved objections and no open or remaining obligations recorded. Verification remains convincing_unverified: no computational evidence or external formal/independent proof verification is claimed. However, stop_reason is recorded as 'unresolved', contrary to the achieved/no-ob"

A.45 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / limitation

"Verification/status caveat: the artifact labels itself an unverified research note and warns that any recorded Python or SageMath execution is not proof by itself. All six reasoning-step reviews are accepted, and global closure is accepted with the stated basis 'objective achieved; no remaining obligations; no unresolved objections'; nevertheless, objective_review/status is 'partial' and coordinator_decision is 'stop_no_progress'. These status differences are not explained in the supplied fields"

A.46 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / limitation

"Verification status remains explicitly 'incomplete', despite acceptance of the displayed argument steps and reports that all five proof obligations are established: convexity/region identification, exact increments, the universal upper bound, an attaining construction, and empty-arrangement/small-value checks. Both unresolved- and resolved-objection lists are empty; the batch does not explain the incomplete status."

A.47 project:01af8ffa-60c3-4eea-bc15-36dca347ddfa, strategy:16d591eb-fd1d-4719-93bc-f2e038a43f85, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 1

"The batch contains the removed strategy's proof but no contents for the session's logged artifact. No explicit positive-n small-value checks are supplied, despite their being requested."

A.48 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 2

"The batch contains no mathematical claim, proof, calculation, counterexample, or test outcome with which to assess the strategy."

A.49 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 3

"The batch states that y = ix + i^2 attains the bound, but the supplied reasoning fields do not include a verification that this construction has the required distinct intersections."

A.50 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 4

"The batch contains only JSON-schema metadata, not substantive research findings, methods, executed tests, or open research questions. Allowed labels such as counterexample, inconclusive, and contradicts do not establish actual outcomes or contradictions."

A.51 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 5

"The batch contains only response-schema definitions and validation constraints, not research findings, executed methods, substantive objections, or open research questions."

A.52 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 6

"The recorded stop_reason 'unresolved' conflicts with the response's statement that no material objection remains and its empty objections list; the batch does not explain the discrepancy."

A.53 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 7

"The asserted exact maximum, complete-argument summary, and acceptance of every listed step conflict with the partial/unresolved outcome and explicit statement that no solution is claimed; the batch does not reconcile these statuses or provide independent verification."

A.54 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 8

"The proof-obligation review marks all listed obligations established and reports no material objection, whereas the conclusion explicitly says the argument remains incomplete and no solution is claimed. The batch does not resolve this status discrepancy or provide independent verification."

A.55 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 9

"The artifact reports a complete deductive solution and skeptic agreement, whereas the session reports partial unresolved progress, incomplete verification, and termination for no progress despite a productive-strategy classification. This is a reporting/status mismatch, not evidence that either independent verification or a mathematical error occurred."

A.56 session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa: evidence index / conflict or gap 10

"The session's five open proof obligations include two displayed obligations described as established; the remaining three obligations are not supplied in this batch."

A.57 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / conflict or gap 11

"The skeptic's stop_reason is 'unresolved', while objective_status is 'achieved', the response says no material objection remains, and objections, unresolved_objections, and remaining_obligations are empty. The batch does not explain this status inconsistency."

A.58 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / conflict or gap 12

"The accepted global closure stating that the objective was achieved differs from the 'partial' objective-review status and 'stop_no_progress' coordinator decision; the batch provides no reconciliation."

A.59 artifact:f60b98af-2fd0-493d-8eb9-1b7b204be8fa:4b50ca5d-ad1f-4f6a-8042-81975427938f: evidence index / conflict or gap 13

"The explicit 'incomplete' verification status is not reconciled with accepted steps, proof obligations reported established, and the strategy review's claim that the maximum is established."

Evidence and authorship note

Snapshot SHA-256: 81d2cc612f3c60017cfe9481c4c910a96a5f4736f6ba901056760ad2cc6064b7 Generated by an AI author from the cited saved research snapshot. This document is not independent mathematical verification. No external bibliography was added.

Validation warnings

- IMPORTANT EVIDENCE COVERAGE LIMITATION: 1068 saved research field fragment(s) were omitted during indexing and were not reviewed for this paper. The synthesis below is provisional and reflects covered evidence only; no conclusion is drawn from omitted material or its absence. Consult the complete frozen evidence snapshot and omission ledger before relying on these findings.
- Author self-review: 'established' marks claims supported by the supplied deductive account; it must not be read as a claim of independent, formal, or computational verification. [C10, C12]
- Author self-review: the record simultaneously reports accepted reasoning and closure while retaining incomplete, partial, unresolved, and stop_no_progress labels. The plan preserves this contradiction rather than silently choosing the favorable interpretation. [C13]
- Author self-review: the five-open-obligations count conflicts with empty obligation lists and reports that all five obligations are established. The missing session entries cannot be reconstructed as though their original text were supplied. [C13, C14]
- Author self-review: an early extract's statement that no response or artifact contents are supplied is local to that extract; later artifact-linked entries do contain mathematical deductions. Conversely, those later summaries do not establish exhaustive access to the originals. [C14]
- Author self-review: the exact increment requires separate treatment of coincident lines and an explanation of why each encountered region splits into exactly two. Omitting either point would leave the universal upper-bound exposition incomplete. Both are included in the claim register and section plan. [C3, C4, C5]
- Author self-review: do not use a recurrence asserting maximal increments before proving attainability. The upper-bound section uses inequalities for arbitrary arrangements, and equality is justified only by the explicit family. [C6, C7, C8, C9]
- Author self-review: no experiments, performance measurements, external citations, failed geometric constructions, or independent replications are supported by the package. None should be added to make the study appear more empirical or extensively validated. [C12, C14]
- Evidence indexing recorded 13 potential conflict or gap note(s); inspect the frozen source records directly.
- Evidence-index self-review (project:01af8ffa-60c3-4eea-bc15-36dca347ddfa, strategy:16d591eb-fd1d-4719-93bc-f2e038a43f85, session:f60b98af-2fd0-493d-8eb9-1b7b204be8fa): The batch contains the removed strategy's proof but no contents for the session's logged artifact. No explicit positive-n small-value checks are supplied, despite their being requested.
- The evidence index explicitly omitted 1137 saved field fragment(s); see the generation coverage ledger and review the complete archived snapshot.
- Long-form authoring used 16 bounded evidence batches covering 5 saved sources. The complete frozen snapshot is archived with this paper; the extracted index may not preserve every source detail.

- The frozen evidence contains open proof/research obligation(s) requiring reconciliation: 5 item(s) are not listed by the author and are not presumed resolved; 4 author-listed item(s) have no exact frozen-record match. See the obligation reconciliation.